PSC 85509:
Applied Quantitative Research II

Instructor: Thomas Leavitt

Official Course Description

This course is the third in the Political Science program's quantitative methods sequence, following PSC 85507 (Quantitative Analysis I: Regression Analysis). The course is intended for students who want to develop the skills to use quantitative methods in their own research and to think systematically about empirical data in political science. The course begins with a review of the foundations of statistical modeling, estimation, and inference, including ordinary least squares (OLS) and maximum likelihood, and then examines how to apply these tools in common empirical settings. Topics include methods for panel, multilevel (hierarchical), event history (survival), and spatial data, as well as the use of these methods in causal inference designs, such as regression discontinuity, difference-in-differences, and instrumental variables.

Students will work with real datasets, replicate and extend published empirical studies, and produce an original empirical research paper applying these methods to a substantive question in political science. The course uses both `R` and `Stata` for statistical analysis. By the end of the semester, students will be able to identify appropriate quantitative methods for different types of data, implement these methods using statistical software, and apply them to conduct and evaluate research in political science.

Course Organization

The course assumes the material of PSC 85507. From that course you should bring descriptive statistics and the normal distribution; bivariate and multiple regression fit by least squares; hypothesis tests, p-values, and confidence intervals; dummy variables, interactions, polynomials, and logarithms as ways of changing a regression's functional form; the omitted-variable-bias formula; heteroskedasticity, autocorrelation, multicollinearity, and influential observations; logit and probit, with their coefficients converted to predicted probabilities, odds ratios, and average marginal effects; and `Stata`. The course assumes no prior experience with `R`.

The course has two parts, each in two halves. Week 1 stands outside the two parts and gives an overview of the course. Part I (Weeks 2–7) builds the framework. Part IA (Weeks 2–5) begins with a population distribution and the estimands defined on it, turns to random sampling from that population, and takes up estimation of and inference about those estimands from a random sample, assuming as little about the population as possible. Part IB (Weeks 6–7) brings regression

into the framework, asking which population quantity ordinary least squares estimates when none of the usual regression assumptions holds. Time permitting, we also touch on maximum likelihood. Part II (Weeks 8–13) adds causal identification. The question is no longer only how to estimate and make inferences about an estimand defined on the population we sample from, but when that estimand equals a causal quantity defined on a counterfactual population from which we cannot sample. Part IIA (Weeks 8–10) rests that equality on mean exchangeability given covariates, the assumption that, among units with the same covariate values, treated and untreated units would have the same average outcome if all were treated and the same average outcome if none were. It then estimates causal quantities through outcome models, propensity score models, and doubly robust combinations of the two. Part IIB (Weeks 11–13) turns to designs that replace mean exchangeability with other assumptions, invoking instruments, thresholds, and timing to identify, estimate, and make inferences about causal effects, typically for a narrower subset of the target population. Week 14 also stands outside the two parts and is for final paper presentations.

The course gives you a framework that outlasts any particular method. New methods keep arriving: Sensitivity analysis for omitted-variable bias, the difference-in-differences estimators surveyed in Week 13, and double machine learning (Chernozhukov et al., 2018; Kennedy, 2024), which lies beyond this course, all emerged within the last decade. Once you have the fundamentals, you can teach yourself whatever comes next.

Course Learning Goals

By the end of this course, students should be able to do the following.

1. Articulate what an estimator targets in a population, and reason carefully about whether the estimator reaches that target under assumptions you are willing to defend.

2. Distinguish the *statistical* question from the *causal* question in any analysis, your own or a published study's, and evaluate the identifying assumptions that let a statistical quantity answer the causal question.

3. Implement the core methods of quantitative political science research in `R` or `Stata` (I provide course code in both languages): plug-in estimators and simulation of their sampling distributions, OLS with sandwich and cluster-robust inference, the bootstrap, outcome-model and propensity-score estimation of causal effects, doubly robust estimation, instrumental variables, regression discontinuity, and difference-in-differences.

4. Read and critically evaluate published quantitative political science research, including identifying threats to the validity of its conclusions and proposing principled extensions.

5. Evaluate a published empirical study's use of a method, and communicate your own analyses through reproducible code, well-formatted tables and figures, and clear written interpretation.

6. Most importantly, learn a new method on your own — from a methods paper, a package vignette, or a replication archive — by asking what the method targets, under what assumptions, and how to quantify the error of its estimator.

Course Assignments

Problem sets (4)

The course has four problem sets, each due at the start of class in the week marked in the schedule below. The four sets follow the course's sequence: Problem set 1 covers the plug-in framework, the population distribution and i.i.d. sampling, unbiasedness and consistency, agnostic inference, the bootstrap, and cluster samples (Weeks 1–5); problem set 2, the CEF, the BLP, and OLS, multiple regression, and nonlinear CEFs (Weeks 6–7); problem set 3, causal identification and estimation through outcome models and propensity-score weighting, with a bonus question on the doubly robust estimator (Weeks 8–10); and problem set 4, instrumental variables, regression discontinuity, and difference-in-differences (Weeks 11–13). I will drop your lowest problem set score, and only the three highest scores count toward your grade. A problem set you do not submit receives a score of zero and counts as your lowest score.

Problem sets will combine analytical questions and replication exercises in `R` and/or `Stata`. Problem sets 3 and 4 add an "Application of Methods" component in which student groups select a published political science article that uses that problem set's methods and evaluate the article's use of those methods. Each group should select a different article; groups coordinate their choices in a shared spreadsheet, first come, first served. Articles covered in lecture or listed in the slide references are not eligible for the Application of Methods component.

On any problem set's due date, I may ask you to complete a short *check of understanding* tied to the problem set just submitted: a brief handwritten answer, *no laptop*, to one of its questions, or a short account of your own submitted code.

Final research paper

Each student will complete an original empirical research paper applying the course's methods to a substantive question of their choosing. The paper should be a polished, professional document suitable as a writing sample for the job market, a draft for journal submission, or a policy report, depending on each student's career objectives.

I strongly encourage students who completed a research paper or replication project in PSC 85507 (or another course) to continue developing that work in PSC 85509, ideally toward a publishable piece. The methods you will learn this term will let you take a paper you wrote with the toolkit of PSC 85507 and deepen the paper's identification strategy, extend the paper to a different data structure, or reframe the paper as a more credible empirical study. The project milestones below give you time to develop the paper substantively.

Project milestones

- **Research proposal** (due Week 7): a 2–3 page document outlining the research question, motivating literature, proposed methodology, and intended data source(s).
- **Draft of final paper** (due at the end of Week 13): a complete first draft of the paper, shared with your discussant on submission.
- **In-class presentation** (Week 14): a 15-minute talk on the paper, with slides. Each student also serves as the discussant for one classmate's paper, having read the classmate's draft,

and delivers 5 minutes of remarks, also with slides. Both slide decks are due by the end of the day before the presentation.

- **Final paper** (due the last day of finals week): the polished final version, incorporating feedback from the draft and presentation.

Grades

Final grades weight the course assessments as follows.

- **30%**: problem sets, the three highest of four scores (**10%** each).
- **5%**: the research paper proposal.
- **10%**: the draft of the final paper.
- **40%**: the final paper.
- **15%**: the in-class final-paper presentation, attendance, and participation (including service as a discussant for classmates' presentations).

Course Readings

This course is ZTC ("Zero Textbook Cost"). I will post all required readings on the course website, and the reading list draws primarily from the following books, all freely available online or through the library. (Freedman, Pisani, and Purves is the one that is not online; it is available through the Mina Rees Library, and I will post the assigned chapters.) I will also upload the dataset for each week's application to the course website, so no assignment requires you to find data.

1. Hansen, Bruce E. (2022). *Probability and Statistics for Economists*. Princeton, NJ: Princeton University Press. (Cited below as "Hansen Vol I.")
2. Hansen, Bruce E. (2022). *Econometrics*. Princeton, NJ: Princeton University Press. (Cited below as "Hansen Vol II.")
3. Aronow, Peter M., and Benjamin T. Miller. (2019). *Foundations of Agnostic Statistics*. New York, NY: Cambridge University Press.
4. Freedman, David, Robert Pisani, and Roger Purves. (2007). *Statistics*. 4th ed. New York, NY: W. W. Norton. (Cited below as "FPP.") Assigned through Week 9 alongside Aronow and Miller and Hansen. It teaches through worked numerical examples and uses no calculus; where a week assigns both, read FPP first and the theory second.
5. Gelman, Andrew, Jennifer Hill, and Aki Vehtari. (2021). *Regression and Other Stories*. New York, NY: Cambridge University Press. The authors post the full text at https://users.aalto.fi/~ave/ROS.pdf, free to view and download for personal use only; that PDF's page numbers match the print edition. Code and data accompany the book at https://avehtari.github.io/ROS-Examples/.

6. Hernán, Miguel A., and James M. Robins. (2020). *Causal Inference: What If*. Boca Raton, FL: Chapman & Hall/CRC. Available online at https://miguelhernan.org/whatifbook. (Page references follow the online edition dated November 21, 2025.)

7. Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik. (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations*. New York, NY: Cambridge University Press. (*Elements in Quantitative and Computational Methods for the Social Sciences*.) Available online at https://arxiv.org/abs/1911.09511. (Week 12's page references follow that version.)

8. Efron, Bradley, and Robert J. Tibshirani. (1993). *An Introduction to the Bootstrap*. Boca Raton, FL: Chapman & Hall/CRC. FPP does not cover the bootstrap; Week 5's treatment of the bootstrap comes from this book.

In addition to texts focused on statistical and causal inference methods, each week's reading list from Week 4 on includes an application of those methods to a substantive empirical question. We will draw applications from American politics, comparative politics, and international relations.

Course Standards

Electronics

To the extent possible, **please bring a personal laptop to each class** for class-related purposes, mostly in-class exercises using statistical software. However, please do not use your cell phones or other electronic devices during class. If a cell phone or another device is important for your participation in the course, please let me know; I will adjust the policy for the entire class rather than single anyone out.

Attendance, Participation, and Collaboration

Attendance and active participation are extremely important to learning from this course. I strongly encourage you to collaborate with other students on problem sets, and the "Application of Methods" components are explicitly group-based. However, all written submissions should be in your own words.

If you need to miss class, please do **not** contact me ahead of time. A missed class affects the participation component of your grade the same way whether or not you contact me beforehand. Please contact me only if you have to miss a class in which you are scheduled to present.

Incomplete and Late Assignments

If you think that you will be unable to complete an assignment by its due date because of extenuating circumstances, please **do** let me know ahead of time. I may change the assignment's deadline for the entire class. If you turn in the final paper late without my prior consent, I will deduct one-third of a letter grade per day. Because I typically review problem sets in class, I will not accept late problem sets for credit.

Use of Artificial Intelligence (AI) Tools

Because this course is fundamentally about quantitative reasoning and statistical computing, I am not forbidding the AI tools that are reshaping how such work is done. You are welcome — encouraged, even — to use large language models (ChatGPT, Claude, Copilot, and the like) to debug your `R` or `Stata` code, decipher cryptic error messages, brainstorm questions worth asking of the data, and draft the routine syntax you would otherwise type out by rote.

A single well-posed prompt already does much of the work. A scripted pipeline does more: It sweeps through thousands of records, merges them with other sources, and logs every step, which ad hoc prompting cannot do at any scale you can audit or reproduce.

That latitude comes with one non-negotiable condition: You own everything you submit.

- **The output is yours.** "The model hallucinated" and "that is just what the chatbot gave me" are never valid defenses. If your code quietly returns the wrong answer, or your paper cites a study that was never written, the error is yours and no one else's. Verify everything these tools produce before your name goes on anything you used the tools for.
- **You must be able to explain it.** You may use AI to help write code, but you are answerable for every line you hand in. If I ask you in class or in office hours to walk me through a particular function or loop in your work and you cannot say how the function or loop behaves or why it is there, that portion of the assignment may receive no credit. More important still, I want you to ask at every step what could be wrong with a given result — what sanity check, cross-tabulation, or quick plot would tell you whether the result is right. Any check of understanding on a problem set's due date asks whether you can explain your own submission, unaided. The ability to explain your own submission is what distinguishes sound use of AI tools.
- **Think before you prompt.** These tools are excellent at carrying out a task you have already framed and poor at supplying the insight that framed it. Do not surrender your judgment to them; use them to move faster on questions that are already your own, not to do your thinking for you.
- **Disclosure.** Any assignment on which you relied on AI tools must include a brief disclosure statement. As journals ask of their authors, it should (i) name the specific tool(s) and the version or model you used (for example, the product name together with its model or version identifier), (ii) state what you used each one for — which tasks, and which parts of the work — and (iii) confirm that you reviewed and verified the output and take full responsibility for it. As in those venues, an AI tool is never a co-author and never answerable for the work; that responsibility is yours alone.

The course's academic-integrity standards apply to AI-assisted work exactly as they do to everything else, and I treat undisclosed use of AI as a breach of them. These tools also routinely misattribute or invent their sources, so catching and correcting such errors before you submit is part of the same responsibility.

Academic Integrity

I will not tolerate violations of academic integrity under any circumstances. Please familiarize yourself with the CUNY Policy on Academic Integrity. If you are having difficulty with an

assignment, do not attempt to present another author's work as your own; please come talk to me instead.

Software and Computing

This course uses both `R` and `Stata`, and the bilingualism is my job, not yours: **All code I provide — in lectures and problem sets — will be available in both languages**, so that the course accommodates every student's choice. **There is no bilingual requirement for any student work.** Submit every assignment — problem sets and the final paper — in whichever language you prefer. PSC 85507 is taught in `Stata`; although nothing you submit has to be in `R`, a goal of this course is for you to leave equally comfortable in `R`.

Why `R` is worth the investment

If you have not yet used `R` seriously, the start of this course will require an investment of time. That investment pays off, for several reasons.

- **Frontier methods are `R`-first.** New causal-inference and machine-learning methods typically appear in `R` years before they reach `Stata`, if they reach it at all. The `did`, `DoubleML`, `HonestDiD`, and `sensemakr` packages reached `Stata` only when other authors later reimplemented the `R` originals; causal forests (`grf`) waited years for a `Stata` counterpart; and at the time of this writing, kernel balancing (`kbal`) has no `Stata` implementation at all. Working at the methodological frontier usually means working in `R`.
- **Free and open source.** `R` runs on any platform, costs nothing, and requires no license and no institutional affiliation. That accessibility matters for your collaborators, for replication archives, and for the work you do after graduate school.
- **The standard in political science methodology.** Replication archives for *Political Analysis*, the *American Journal of Political Science*, the *Journal of Politics*, and similar venues now overwhelmingly ship `R` code. You need `R` to read recent methods papers in political science.
- **Reproducible research workflows.** `R Markdown` and `Quarto` integrate code, output, citations, and prose into a single document, so you can write a polished, fully reproducible paper or report without ever leaving that document.
- **Visualization.** The `ggplot2` package gives you finer control over a figure than `Stata`'s graphics commands do. Political science journals increasingly publish figures that authors build in `ggplot2`.
- **Job market.** Data-science, government-analyst, and industry positions often expect `R` or `Python` (or both). `Stata` alone narrows your opportunities outside academic econometrics. Of the software skills you can build in this course, `R` repays you the most.

Installing the software

To download `R`, please go to https://cloud.r-project.org and follow the installation instructions. After installing `R`, please download and install `RStudio` from https://posit.co/

download/rstudio-desktop/. CUNY provides access to Stata, which you can download for your personal computer; Stata is also installed on lab computers at the Graduate Center.

A valuable free resource for learning R is Wickham et al. (2023), available at https://r4ds.hadley.nz/.

Recommended R preparation on DataCamp

I have set up a free classroom for this course on DataCamp, an online service that teaches coding through short exercises you complete in a web browser. The classroom holds twenty-two chapters, sequenced to follow the course. **Every chapter is recommended. No chapter is required or graded.** Each chapter carries a date in DataCamp, set to the night before that week's class. Those dates mark a suggested pace.

Suggested	*DataCamp course*	*Chapters*
Before Week 2	Introduction to R	Intro to basics, Vectors, Matrices, Factors, Data frames, Lists
Before Week 3	Intermediate R	Conditionals and Control Flow, Loops, Functions, The apply family
Before Week 3	Intermediate Importing Data in R	Importing data from statistical software packages
Before Week 4	Introduction to the Tidyverse	Data wrangling
Before Week 5	Introduction to the Tidyverse	Data visualization
Before Week 6	Introduction to the Tidyverse	Grouping and summarizing
Before Week 7	Introduction to the Tidyverse	Types of visualizations
Before Week 8	Introduction to Data Visualization with ggplot2	Introduction
Before Week 9	Introduction to Data Visualization with ggplot2	Aesthetics
Before Week 10	Introduction to Data Visualization with ggplot2	Geometries
Before Week 11	Introduction to Data Visualization with ggplot2	Themes
Before Week 12	Reshaping Data with tidyr	From Wide to Long and Back
Before Week 13	Joining Data with dplyr	Joining Tables
Before Week 14	Joining Data with dplyr	Left and Right Joins

Course Schedule and Readings

The CUNY Graduate Center's academic calendar is available via the Office of the Registrar here.

Within a week, anything **due** comes first, marked ⋆ and set in **blue**; anything distributed that week follows, marked ∘.

Week 1: Course overview

No assigned readings.

Part I (Weeks 2–7): The framework

Part IA (Weeks 2–5): Estimands, sampling, and inference

Week 2: The population distribution, estimands, and i.i.d. sampling

Required readings

Theory

- FPP Ch. 13, *What Are the Chances?* (pp. 221–236, conditional probabilities, the multiplication rule, independence)
- FPP Ch. 16, *The Law of Averages* (pp. 273–287); Ch. 17, *The Expected Value and Standard Error*, §§1–2 (pp. 288–294)
- Aronow and Miller (2019) §1.1 (pp. 4–15, random events, joint and conditional probabilities, independence), §§1.2.1–1.2.3 (pp. 16–24, random variables and cumulative distribution functions)
- Aronow and Miller (2019) §§2.1.1–2.1.3 (pp. 45–59, expected values, variances, mean squared error), §3.1 (pp. 91–96, i.i.d. random variables and random sampling)

Recommended readings

- Hansen Vol I Ch. 2 (pp. 24–56), §§6.4–6.11 (pp. 130–137)
- Wickham et al. (2023): available here

Week 3: The plug-in principle, unbiasedness, the standard error, and consistency — the gap between estimator and estimand, and how fast that gap shrinks

- Problem set 1 distributed

Required readings

Theory

- FPP Ch. 4, *The Average and the Standard Deviation* (pp. 57–77); Ch. 17, *The Expected Value and Standard Error* (pp. 288–307), in full; Ch. 20, *Chance Errors in Sampling* (pp. 355–374)
- Aronow and Miller (2019) §3.2.1 (pp. 96–102, convergence in probability, the continuous mapping theorem, the weak law of large numbers), §3.2.2 (pp. 102–105, bias, consistency, mean squared error)
- Aronow and Miller (2019) §3.3 (pp. 116–121, the plug-in principle and its regularity conditions)

Recommended readings

- Hansen Vol I §7.3 (p. 149), §7.5 (p. 152), §7.10 (p. 155)

Week 4: Agnostic inference — the central limit theorem, estimating standard errors, confidence intervals, and tests

Required readings

Theory (asymptotic normality)

- FPP Ch. 5, *The Normal Approximation for Data* (pp. 78–96); Ch. 18, *The Normal Approximation for Probability Histograms* (pp. 308–330)
- Aronow and Miller (2019) §3.2.4 (pp. 108–111, convergence in distribution, the central limit theorem, Slutsky's theorem), §3.2.5 (pp. 111–114, asymptotic normality)

Theory (estimating standard errors, confidence intervals, tests)

- FPP Ch. 21, *The Accuracy of Percentages* (pp. 375–394); Ch. 23, *The Accuracy of Averages* (pp. 409–437); Ch. 26, *Tests of Significance* (pp. 475–500); Ch. 29, *A Closer Look at Tests of Significance*, §§1–5 (pp. 545–562)
- Aronow and Miller (2019) §3.2.3 (pp. 105–108, the plug-in sample variance, the unbiased sample variance), §3.2.6 (pp. 114–120), §§3.4.1–3.4.2 (pp. 124–130)
- Hansen Vol II §§4.15–4.16 (pp. 111–114, the sandwich)

Application

- Gilens and Page (2014)

Recommended readings

- Hansen Vol I §8.11 (p. 173, the asymptotic distribution of a plug-in estimator), §8.2 (p. 165), §8.6 (p. 170), §8.9 (p. 172, the delta method)
- Hansen Vol II §7.3 (p. 164, asymptotic normality of OLS)
- Gelman et al. (2021, §§4.4–4.5, pp. 57–62, Type S and Type M error, the significance filter, forking paths)

Week 5: The bootstrap and cluster samples — inference when there is no formula, and when the draws are not independent

Required readings

Theory (bootstrap)

- Aronow and Miller (2019) §3.4.3 (pp. 130–132, the bootstrap), §3.4.4 (pp. 132–135, small samples)
- Efron and Tibshirani (1993, Chs. 1–2)
- Hansen Vol II §10.29 (p. 299) and §10.30 (p. 301, wild and cluster bootstrap)

Theory (cluster samples)

- FPP Ch. 19, *Sample Surveys*, §§4–8 (pp. 339–354, probability methods, the Gallup poll, chance error and bias); Ch. 22, *Measuring Employment and Unemployment* (pp. 395–407, the Current Population Survey's clustered design and its weighting)

- Aronow and Miller (2019) §3.5 (pp. 135–141, estimation and inference with clustering)
- Hansen Vol II §§4.22–4.23 (pp. 123–128, cluster-robust), §4.24 (pp. 128–130, at what level to cluster)

Application

- Gilens and Page (2014), continued

Recommended readings

- Cameron and Miller (2015)
- Cameron et al. (2008)
- Efron and Tibshirani (1993, Chs. 3–7)

Part IB (Weeks 6–7): Regression as a plug-in

Week 6: The CEF, the BLP, and OLS as the BLP's plug-in

⋆ **Problem set 1 due**

○ Problem set 2 distributed

Required readings

Theory

- FPP Chs. 8–12 (pp. 119–216): *Correlation*; *More about Correlation*; *Regression*; *The R.M.S. Error for Regression*; *The Regression Line*
- Aronow and Miller (2019) §§2.2.3–2.2.5 (pp. 67–84), §4.1 (pp. 143–151) — read §§4.1.1–4.1.2 (pp. 144–145) before §4.1.3 (p. 147)

Application

- Mincer (1974), used throughout Hansen Vol II Chs. 2–4

Recommended readings

- Hansen Vol II §§2.3–2.18 (pp. 16–40)

Week 7: Multiple regression and nonlinear CEFs

⋆ **Research proposal due**

Required readings

Theory (adding an X: short and long regressions)

- FPP §12.3, *Does the Regression Make Sense?* (pp. 211–213), re-read from Week 6
- Hansen Vol II §2.24 (pp. 45–46)

Theory (multiple regression and nonlinear CEFs)

- Gelman et al. (2021, Ch. 10, pp. 131–152, linear regression with multiple predictors)
- Aronow and Miller (2019) §4.2 (pp. 151–156, inference for regression)
- Aronow and Miller (2019) §4.3 (pp. 156–170, polynomials, overfitting, interactions)

Application

- Mincer (1974); Aronow and Miller (2019) §4.4 (pp. 170–177, clean water and infant mortality)

Recommended readings

- Hansen Vol II §§2.3–2.18 (pp. 16–40), Ch. 4 (pp. 97–136)
- Gelman et al. (2021, Ch. 12, pp. 183–214, transformations and regression)

Part II (Weeks 8–13): Causal identification

Part IIA (Weeks 8–10): Identification from observed covariates

Week 8: Causal estimands, potential outcomes, and the counterfactual and observable distributions

⋆ **Problem set 2 due**

○ Problem set 3 distributed

Required readings

Theory

- FPP Ch. 1, *Controlled Experiments* (pp. 3–11, a treatment group and a comparison group)
- Aronow and Miller (2019) §7.1 (pp. 236–256), this week §7.1.1 (pp. 236–238, the potential outcomes model; the average treatment effect as a difference of two averages) and §7.1.2 (pp. 238–239, what the observed data show of each potential outcome: Example 7.1.5's half-blank table); §§7.1.3–7.1.7 are Weeks 9 and 10
- Hernán and Robins (2020, Ch. 1, pp. 3–12, the definition of a causal effect; causation against association, Figure 1.1)

Application

- Blattman (2009)

Recommended readings

- Holland (1986)
- Imbens and Rubin (2015, Chs. 1–3)
- Gelman et al. (2021, Ch. 18, pp. 339–362; identification at p. 351, the stable unit treatment value assumption (SUTVA) at pp. 352–354)

Week 9: The difference in means, mean exchangeability, and inference

Required readings

Theory

- FPP Ch. 2, *Observational Studies* (pp. 12–29, confounding; the weighted average of pp. 19–20 is the comparison within groups); Ch. 27, *More Tests for Averages*, §§1–4 (pp. 501–517, the standard error for a difference, used in this week's class and Problem set 3; experiments)
- Aronow and Miller (2019) §7.1, continued from Week 8: §7.1.4 (pp. 244–247, independence and positivity; the difference in means, Theorem 7.1.10; the selection-bias decomposition, p. 247) and §7.1.5 (pp. 247–250, strong ignorability, which this course calls mean exchangeability given the covariates, with overlap; the comparison within groups, Theorems 7.1.13–7.1.14)
- Hernán and Robins (2020, Chs. 2–3, pp. 13–46; exchangeability at p. 14, conditional exchangeability and standardization at pp. 17–20, the identifiability conditions and positivity at pp. 27–32)

Application

- Blattman (2009), continued

Recommended readings

- Aronow and Miller (2019) §7.1.3 (pp. 240–244, bounds on the average treatment effect with no assumption) and §7.1.7 (pp. 252–254, post-treatment variables), both in the appendix of this week's slides
- Cinelli and Hazlett (2020, §3.1, pp. 43–45, omitted variable bias; §4.3, pp. 50–52, the robustness value; §4.4, pp. 52–54, benchmarking; sensitivity analysis is named in this week's appendix and not taught)
- Hazlett (2020)

Week 10: Outcome models and propensity score models

Required readings

Theory (outcome models)

- Aronow and Miller (2019) §7.2.1 (pp. 256–258, the within-group plug-in estimator) and §7.2.2 (pp. 258–261, the regression estimator, Definition 7.2.3: predict both potential outcomes and average)

Theory (propensity score models and weighting)

- Aronow and Miller (2019) §7.1.6 (pp. 250–252, the propensity score and balance, Theorem 7.1.16), §7.2.5 (p. 264, the propensity score estimated by a logit), §7.2.6 (pp. 264–267, weighting by the inverse of the propensity score: the Horvitz–Thompson and Hájek estimators; inference by the bootstrap), §7.2.7 (pp. 267–270, doubly robust estimators), §7.3 (pp. 271–276, overlap and positivity)

Application

  - Blattman (2009), continued (pp. 234–236, the argument that the chance of abduction varied with age and place alone)

Recommended readings

- Aronow and Miller (2019) §7.2.3 (p. 261, plug-ins fit by maximum likelihood), §7.2.4 (pp. 262–264, matching), §7.2.8 (pp. 270–271, placebo tests), and §7.5 (pp. 276–280, the authors' own application); none of the four is taught in class
- Rosenbaum and Rubin (1983, pp. 41–45; the propensity score named, balance (Theorem 1), and identification given the score (Theorem 3))
- Angrist (1998, §2, the weights a regression's coefficient on the treatment puts on the groups' comparisons)
- Gelman et al. (2021, Ch. 19, pp. 363–382, causal inference using regression on the treatment variable)
- Hernán and Robins (2020, Ch. 11, pp. 153–162, why model; §§13.1–13.3, pp. 175–179, standardization by predicting each potential outcome; Technical Point 13.2, p. 184, doubly robust estimation; §§15.1–15.2, pp. 199–203, outcome regression; the propensity score from a logistic model)
- Hansen Vol II §2.30 (pp. 50–55, causal effects and the CEF)
- Gelman et al. (2021, Chs. 13–14, pp. 217–262, logistic regression, the usual model for a propensity score)
- Gelman et al. (2021, Ch. 20, pp. 399–413, the propensity-score workflow)
- Hernán and Robins (2020, Ch. 12, inverse-probability weighting)

Part IIB (Weeks 11–13): Identification from instruments, thresholds, and timing

Week 11: Instrumental variables — the effect for compliers

⋆ **Problem set 3 due**

○ Problem set 4 distributed

Required readings

Theory

- Angrist et al. (1996)
- Hansen Vol II §§12.1–12.12 (pp. 337–352), §12.34 (p. 388, the local average treatment effect), §12.36 (p. 392), §12.38 (p. 399, weak instruments)

Application

- Acemoglu et al. (2001)

Recommended readings

- Hernán and Robins (2020, Ch. 16, pp. 209–226)
- Sovey and Green (2011)
- Gelman et al. (2021, §21.1, pp. 421–432, the complier average causal effect)

Week 12: Regression discontinuity — the effect at the cutoff

Required readings

Theory

- Cattaneo et al. (2020, Ch. 2, pp. 8–19; Ch. 3, pp. 20–38; Ch. 4, pp. 39–87; Ch. 5, pp. 88–108)

Application

- Lee (2008)

Recommended readings

- Lee and Lemieux (2010)
- Hansen Vol II Ch. 21 (pp. 755–767)
- Caughey and Sekhon (2011)
- Eggers et al. (2015)
- Gelman et al. (2021, §21.3, pp. 432–440, fuzzy regression discontinuity as an IV problem)

Week 13: Difference-in-differences on panel data — the effect for the treated

⋆ **Draft of final paper due at the end of the week**

Required readings

Theory (panel structure)

- Hansen Vol II §§17.1–17.13 (pp. 609–625), §§17.24–17.25 (p. 637)
- Beck and Katz (1995)

Theory (difference-in-differences)

- Hansen Vol II Ch. 18 (pp. 664–676)
- Roth et al. (2023)

Application

- Paglayan (2019)

Recommended readings

- Goodman-Bacon (2021)
- Callaway and Sant'Anna (2021)
- Ward (2020)
- Gelman et al. (2021, §21.4, pp. 442–445, two-group difference-in-differences and parallel trends)

Week 14: Final paper presentations

- ⋆ **Presentation and discussant slides due the day before the presentations**
- ⋆ **Problem set 4 due**
- ⋆ **Final paper due the last day of finals week**

  A 15-minute talk on your final paper, plus 5 minutes of discussant remarks on one classmate's paper; slides for both are due by the end of the day before, so the presentations run back-to-back without setup time and fill the whole session.

  Last 15 minutes: **course evaluations**. Bring a device you can complete them on.

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